# FUNCTORIALITY OF THE IDENTIFICATION OF THE COHOMOLOGY OF O_{P^N_R}(D): THE IDENTIFICATIONS OF EQUATION (IDENTIFY) ARE COMPATIBLE WITH BASE CHANGE ALONG RING MAPS R -> R', AND FOR F IN R[T_0,...,T_N] HOMOGENEOUS OF DEGREE M THE MAP ON COHOMOLOGY INDUCED BY MULTIPLICATION BY F : O(D) -> O(D+M) IS MULTIPLICATION BY F ON H^0 AND THE CONTRAGREDIENT OF MULTIPLICATION BY F FROM (R[T])_{-N-1-(D+M)} TO (R[T])_{-N-1-D} ON H^N.

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2101-2121 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 2123-2212, the complete author proof (88 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Full derivation in span; the parenthetical basis-element verification is written by the author, not omitted.

*Editorial note.* Span computed from the \label marker through the enclosing \end{lemma}; precard_gate RESULT=OK (probe L9999).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

*Source reference.* Original lemma 01FD: sources/stacks-coherent/pinned-original-source.tex, source lines 2212–2212. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01XU: sources/stacks-coherent/pinned-original-source.tex, source lines 2089–2097. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01XT: sources/stacks-coherent/pinned-original-source.tex, source lines 1820–1823. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *The identifications of Equation (01XU) are compatible with base change w.r.t. ring maps $R \to R'$. Moreover, for any $f \in R[T_0, \ldots, T_n]$ homogeneous of degree $m$ the map multiplication by $f$*

$$\mathcal{O}_{\mathbf{P}^n_R}(d) \longrightarrow \mathcal{O}_{\mathbf{P}^n_R}(d+m)$$

*induces the map on the cohomology group via the identifications of Equation (01XU) which is multiplication by $f$ for $H^0$ and the contragredient of multiplication by $f$*

$$(R[T_0, \ldots, T_n])_{-n-1-(d+m)} \longrightarrow (R[T_0, \ldots, T_n])_{-n-1-d}$$

*on $H^n$.*

*Proof.* Suppose that $R \to R'$ is a ring map. Let $\mathcal{U}$ be the standard affine open covering of $\mathbf{P}^n_R$, and let $\mathcal{U}'$ be the standard affine open covering of $\mathbf{P}^n_{R'}$. Note that $\mathcal{U}'$ is the pullback of the covering $\mathcal{U}$ under the canonical morphism $\mathbf{P}^n_{R'} \to \mathbf{P}^n_R$. Hence there is a map of Čech complexes

$$\gamma : \check{\mathcal{C}}^\bullet_{ord}(\mathcal{U}, \mathcal{O}_{\mathbf{P}_R}(d)) \longrightarrow \check{\mathcal{C}}^\bullet_{ord}(\mathcal{U}', \mathcal{O}_{\mathbf{P}_{R'}}(d))$$

which is compatible with the map on cohomology by Cohomology, Lemma 01FD. It is clear from the computations in the proof of Lemma 01XT that this map of Čech complexes is compatible with the identifications of the cohomology groups in question. (Namely the basis elements for the Čech complex over $R$ simply map to the corresponding basis elements for the Čech complex over $R'$.) Whence the first statement of the lemma.

Now fix the ring $R$ and consider two homogeneous polynomials $f, g \in R[T_0, \ldots, T_n]$ both of the same degree $m$. Since cohomology is an additive functor, it is clear that the map induced by multiplication by $f + g$ is the same as the sum of the maps induced by multiplication by $f$ and the map induced by multiplication by $g$. Moreover, since cohomology is a functor, a similar result holds for multiplication by a product $fg$ where $f, g$ are both homogeneous (but not necessarily of the same degree). Hence to verify the second statement of the lemma it suffices to prove this when $f = x \in R$ or when $f = T_i$. In the case of multiplication by an element $x \in R$ the result follows since every cohomology groups or complex in sight has the structure of an $R$-module or complex of $R$-modules. Finally, we consider the case of multiplication by $T_i$ as a $\mathcal{O}_{\mathbf{P}^n_R}$-linear map

$$\mathcal{O}_{\mathbf{P}^n_R}(d) \longrightarrow \mathcal{O}_{\mathbf{P}^n_R}(d+1)$$

The statement on $H^0$ is clear. For the statement on $H^n$ consider multiplication by $T_i$ as a map on Čech complexes

$$\check{\mathcal{C}}^\bullet_{ord}(\mathcal{U}, \mathcal{O}_{\mathbf{P}_R}(d)) \longrightarrow \check{\mathcal{C}}^\bullet_{ord}(\mathcal{U}, \mathcal{O}_{\mathbf{P}_R}(d+1))$$

We are going to use the notation introduced in the proof of Lemma 01XT. We consider the effect of multiplication by $T_i$ in terms of the decompositions

$$\check{\mathcal{C}}^\bullet_{ord}(\mathcal{U}, \mathcal{O}_{\mathbf{P}_R}(d)) = \bigoplus\nolimits_{\vec{e} \in \mathbf{Z}^{n+1},\ \sum e_i = d} \check{\mathcal{C}}^\bullet(\vec{e})$$

and

$$\check{\mathcal{C}}^\bullet_{ord}(\mathcal{U}, \mathcal{O}_{\mathbf{P}_R}(d+1)) = \bigoplus\nolimits_{\vec{e} \in \mathbf{Z}^{n+1},\ \sum e_i = d+1} \check{\mathcal{C}}^\bullet(\vec{e})$$

It is clear that it maps the subcomplex $\check{\mathcal{C}}^\bullet(\vec{e})$ to the subcomplex $\check{\mathcal{C}}^\bullet(\vec{e}+\vec{b}_i)$ where $\vec{b}_i = (0, \ldots, 0, 1, 0, \ldots, 0))$ the $i$th basis vector. In other words, it maps the summand of $H^n$ corresponding to $\vec{e}$ with $e_i < 0$ and $\sum e_i = d$ to the summand of $H^n$ corresponding to $\vec{e} + \vec{b}_i$ (which is zero if $e_i + b_i \geq 0$). It is easy to see that this corresponds exactly to the action of the contragredient of multiplication by $T_i$ as a map

$$(R[T_0, \ldots, T_n])_{-n-1-(d+1)} \longrightarrow (R[T_0, \ldots, T_n])_{-n-1-d}$$

This proves the lemma. □